Vine space simplifies matrix completion

Incomplete Pearson matrix

	X_1	X_2	X_3	X_4	X_5
X_1	1.00	−0.59	?	0.31	−0.37
X_2	−0.59	1.00	0.54	−0.79	?
X_3	?	0.54	1.00	−0.06	?
X_4	0.31	−0.79	−0.06	1.00	?
X_5	−0.37	?	?	?	1.00

Vine construction: actuarial judgement

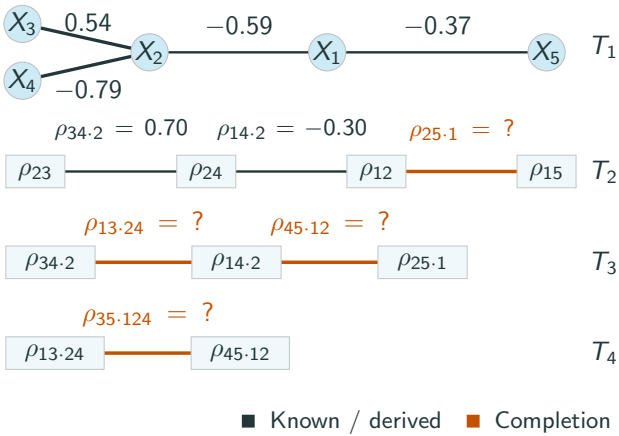
Greedy: strongest $|\rho|$ first

Rules: proximity; connected; acyclic

T_1 edges become T_2 nodes

Partial correlation: $\rho_{ij\cdot k} = \dfrac{\rho_{ij} - \rho_{ik}\rho_{jk}}{\sqrt{(1-\rho_{ik}^2)(1-\rho_{jk}^2)}}$

Partial-correlation vine



■ Known / derived ■ Completion

Choosing partial correlations completes the matrix

Assign partial correlations

Actuarial judgement: assign $\rho_{ij \cdot S}$

Maximise $\det R = \prod_{e \in E(V)} (1 - \rho_e^2)$

$\rho_{25 \cdot 1} = 0.00$ $\quad$ $\rho_{13 \cdot 24} = 0.00$

$\rho_{45 \cdot 12} = 0.00$ $\quad$ $\rho_{35 \cdot 124} = 0.00$

Worked example

$\rho_{25} = \rho_{12}\rho_{15} + \rho_{25 \cdot 1}\sqrt{(1 - \rho_{12}^2)(1 - \rho_{15}^2)}$

$\rho_{25 \cdot 1} = 0.00 \implies \rho_{25} = \rho_{12}\rho_{15} \approx 0.22$

Completed Pearson matrix

	X_1	X_2	X_3	X_4	X_5
X_1	1.00	−0.59	−0.46	0.31	−0.37
X_2	−0.59	1.00	0.54	−0.79	0.22
X_3	−0.46	0.54	1.00	−0.06	0.17
X_4	0.31	−0.79	−0.06	1.00	−0.12
X_5	−0.37	0.22	0.17	−0.12	1.00

Vine is a language to understand dependence and apply our understanding

Code · notebook · demo